Plant Disease Detection System for Sustainable Agriculture

A Project Report

submitted in partial fulfillment of the requirements

of

AICTE Internship on AI: Transformative Learning
with
TechSaksham – A joint CSR initiative of Microsoft & SAP

by

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Under the Guidance of

P. Raja

ACKNOWLEDGEMENT

I would like to express sincere thanks to our Supervisor **P. Raja**, for his admirable guidance and inspiration both theoretically and practically and most importantly for the drive to complete project successfully. Working under such an eminent guide was our privilege.

I also express my sincere thanks to **Edunet** for providing this wonderful opportunity through its internship program. The experience has been incredibly enriching, offering us a platform to learn and grow during the course of the program.

I express thanks to **my parents** for their love, care and moral support without which we would have not been able to complete this project. It has been a constant source of inspiration for all our academic endeavours. Last but not the least, I thank the **Almighty** for making us a part of the world.

ABSTRACT

The primary occupation in India is agriculture. India ranks second in the agricultural output worldwide. Here in India, farmers cultivate a great diversity of crops. Various factors such as climatic conditions, soil conditions, various disease, etc affect the production of the crops. The existing method for plants disease detection is simply naked eye observation which requires more man labour, properly equipped laboratories, expensive devices, etc. And improper disease detection may lead to inexperienced pesticide usage that can cause development of long-term resistance of the pathogens, reducing the ability of the crop to fight back. The plant disease detection can be done by observing the spot on the leaves of the affected plant. The method we are adopting to detect plant diseases is image processing using Convolution neural network (CNN). Thus, accurate disease prediction is one of the greatest challenges, despite many advances in agriculture department in recent decades. Plants means “Food” and Food means “Life”. Plant Disease Detection is closely related to agriculture sector, which contributes significantly to the economy of the nation. The proposed system helps in identification of plant disease and provides remedies. It can be used as a defence mechanism against the disease. The database obtained from the Internet is properly segregated and the different plant species are identified and are renamed to form a proper database then obtain test-database which consists of various plant diseases that are used for checking the accuracy and confidence level of the project .Then using training data we will train our classifier and then output will be predicted with optimum accuracy . We use Convolution Neural Network (CNN) which comprises of different layers which are used for prediction. With our code and training model we have achieved an accuracy level of 97%. Oursoftware gives us the name of the plant species with its confidence level.

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CHAPTER 1

Introduction

1.1Problem Statement:

The problem addressed is the early detection of plant diseases, which significantly impact crop yields and global food security. Traditional methods of disease diagnosis are often time-consuming, subjective, and inaccessible to many farmers. Addressing this issue is critical to reducing crop losses, minimizing pesticide use, and ensuring sustainable agricultural practices.

1.2Motivation:

This project was chosen to leverage AI for improving agricultural productivity by enabling early, accurate plant disease detection. Its potential applications include real-time disease monitoring through mobile apps, with a significant impact on reducing crop losses, optimizing pesticide use, and promoting sustainable farming.

1.3Objective:

The objectives of this project are to develop a reliable AI-based system for accurate detection and classification of plant diseases using leaf images and to provide farmers with a user-friendly tool for early disease diagnosis to improve crop health and sustainability.

1.4Scope of the Project:

Through this project work, we try to detect the plant disease using the pre-determined dataset containing the images of a variety of plant leaves. The model constructed in this project can show an accuracy up to 93% in detecting the disease

CHAPTER 2

Literature Survey

Authors: - Sachin D. Khirade.
Paper: - A survey on Plant Disease Detection using Convolutional neural network. International Journal of Computer Applications, 72(16).

- Identification of the plant diseases is the key to preventing the losses in the yield and quantity of the agricultural product.

- It requires tremendous amount of work, expertise in the plant diseases, and also require the excessive processing time.

- Hence, image processing is used for the detection of plant diseases. Disease detection involves the steps like image acquisition, image pre-processing, image segmentation, feature extraction and classification.

- This paper discussed the methods used for the detection of plant diseases using their leaves images. This paper discussed various techniques to segment the disease part of the plant.

- This paper also discussed some Feature extraction and classification techniques to extract the features of infected leaf and the classification of plant diseases.

- The accurately detection and classification of the plant disease is very important for the successful cultivation of crop and this can be done using image processing.

- This paper discussed various techniques to segment the disease part of the plant. This paper also discussed some Feature extraction and classification techniques to extract the features of infected leaf and the classification of plant diseases.

- The use of ANN methods for classification of disease in plants such as self- organizing feature map, back propagation algorithm, SVMs etc. can be efficiently used. From these methods, we can accurately identify and classify various plant diseases using image processing technique.

Authors:- Amandeep Singh.

- The most significant challenge faced during the work was capturing the quality images with maximum detail of the leaf colour.

- It is very typical task to get the image with all the details within a processable memory.

- Such images are formed a through high resolution and thus are of 6-10MB of size. This was handled by using a Nikon made D5200 camera which served the task very well.

- Second challenge faced was to get rid of illumination conditions as from the start to the end of paddy crop season, illumination varies a lot even when the image acquiring time is fixed.

- However, the solution to this is variable user defined thresholding and making necessary adjustments to the shades of LCC.

Authors: - Satish Madhgoria.

- Proposed automatic pixel-based classification method for detecting unhealthy regions in leaf images is presented.

- The algorithms have been tested extensively. Linear SVM has been used to classify each pixel. We have also shown hoe the results from SVM could be improved remarkably using the neighbourhood check technique.

- The presented algorithm could well extend for other detection tasks which also mainly rely on colour information, but extension to other features is easily possible.

- The task is performed in three steps. First, we perform segmentation to divide the image into foreground and background. In the second step, support vector machines are applied to predict the class of each pixel belonging to the foreground.

- And finally, we do further refinement by neighbourhood-check to omit all falsely-classified pixels from second step. Satish Madhgoria, MarekSchikora & et at. Proposed automatic pixel-based classification method for detecting unhealthy regions in leaf images is presented

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CHAPTER 3

Proposed Methodology

3.1 System Design

Diagram of the Proposed Solution

Below is the proposed system architecture for the Plant Disease Detection System:

1. **Image Input**:
   - Farmers capture images of plant leaves using a mobile device or upload pre-existing images via the system interface.
2. **Preprocessing Module**:
   - Uploaded images are resized, normalized, and augmented to enhance the data quality and adapt it to the model's input size.
3. **Deep Learning Model (CNN-based)**:
   - The preprocessed images are passed through a Convolutional Neural Network (e.g., ResNet, MobileNet) trained to classify plant diseases.
   - The model outputs probabilities for each disease class, indicating the likelihood of a disease.
4. **Postprocessing and Visualization**:
   - The system maps the predicted disease class to user-friendly results, including disease name, severity level, and treatment suggestions.
5. **Result Delivery**:
   - The system displays results via a web or mobile application, providing actionable insights to the user.

Explanation of the Diagram

The design emphasizes modularity and scalability. The image preprocessing ensures data quality, while the CNN model provides accurate classification. The visualization and result delivery focus on usability, allowing farmers to act based on clear and concise information. The model can be deployed on cloud platforms for remote accessibility, and a local deployment option ensures utility in areas with limited internet access.

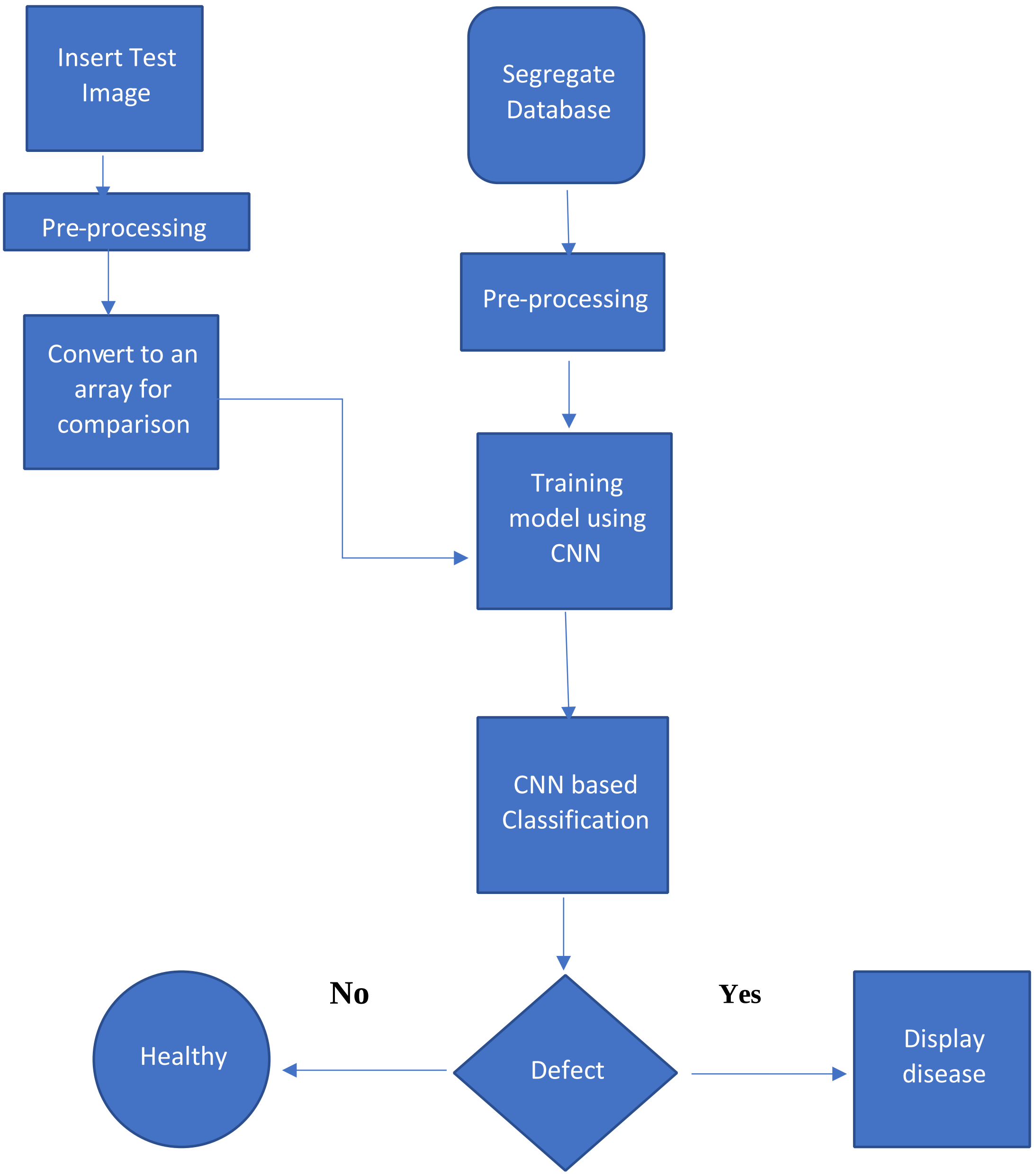


Figure 1: CNN flow chart

3.2 Requirement Specification

Tools and Technologies Required

Hardware Requirements:

- Processor: Intel i5/i7 or equivalent with GPU support for local model training.
- RAM: Minimum 16 GB for efficient processing.
- Storage: 1 TB HDD/SSD for dataset storage and model files.
- Graphics Card: NVIDIA GPU (e.g., GTX 1660 or higher) for accelerated deep learning tasks.
- Smartphone or camera: For capturing high-resolution images of plant leaves.

Software Requirements:

- **Operating System**: Windows 10, macOS, or Linux.
- **Programming Language**: Python (preferred for deep learning frameworks).
- **Deep Learning Frameworks**: TensorFlow, PyTorch, or Keras.
- **Image Processing Libraries**: OpenCV, PIL.
- **Integrated Development Environment (IDE)**: Jupyter Notebook, VS Code, or PyCharm.
- **Web/Mobile Frameworks**: Flask/Django for web deployment; React Native/Flutter for mobile applications.
- **Database**: MySQL or MongoDB for storing user data and disease information.
- **Cloud Platforms (optional)**: AWS, Google Cloud, or Azure for model deployment and scalability.

By integrating these components, the system ensures efficiency, scalability, and user-friendliness. Let me know if you'd like a visual representation of the system architecture!

NEURAL NETWORKS :

Neural networks are multi-layer networks of neurons (the blue and magenta nodes in the chart below) that we use to classify things, make predictions, etc. Below is the diagram of a simple neural network with five inputs, 5 outputs, and two hidden layers of neurons.

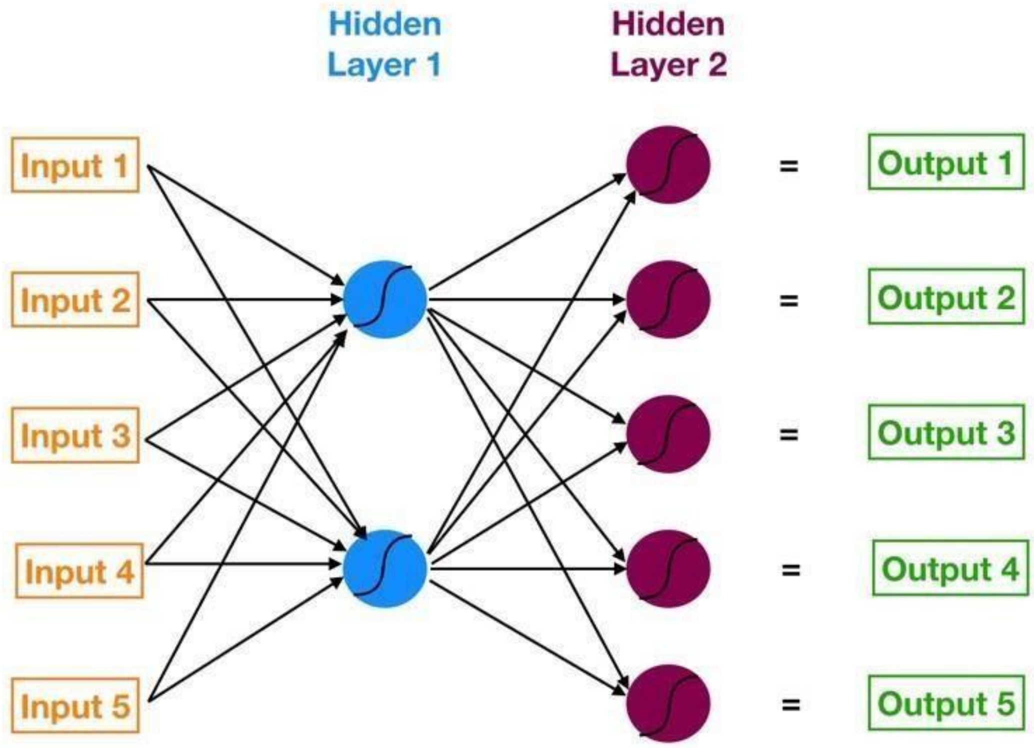


Figure 2: The Layers in the Neural Network

The arrows that connect the dots in the above figure shows how all the neurons are interconnected and how data travels from the input layer all the way through to the output layer.

NEURONS: An artificial neuron (also referred to as a perceptron) is a mathematical function. It takes one or more inputs that are multiplied by values called “weights” and added together. This value is then passed to a non-linear function, known as an activation function, to become the neuron’s output.

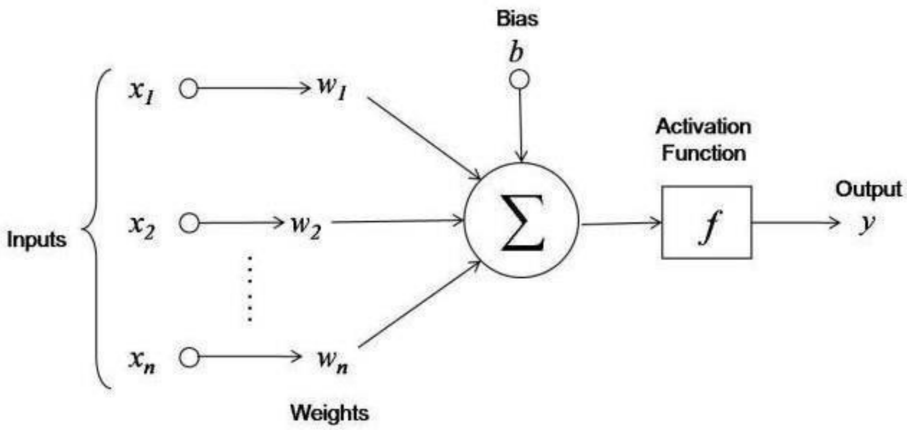


Figure 3: The internal structure of neural network with bias and activation function

- First, it adds up the value of every neurons from the previous column it is connected to. In the above figure, there are 3 inputs (x1, x2, x3) coming to the neuron, so 3 neurons of the previous column are connected to our neuron.
- This value is multiplied, before being added, by another variable called “weight” (w1, w2, w3) which determines the connection between the two neurons. Each connection of neurons has its own weight, and those are the only values that will be modified during the learning process.
- Moreover, a bias value may be added to the total value calculated. It is not a value coming from a specific neuron and is chosen before the learning phase, but can be useful for the network.
- After all those summations, the neuron finally applies a function called “activation function” to the obtained value.
- The so-called activation function usually serves to turn the total value calculated before to a number between 0 and 1 (done for example by a sigmoid function shown by Figure 3). Other function exist and may change the limits of our function, but keeps the same aim of limiting the value.

- That’s all a neuron does! Take all values from connected neurons multiplied by their respective weight, add them, and apply an activation function. Then, the neuron is ready to send its new value to other neurons.

- After every neurons of a column did it, the neural network passes to the next column. In the end, the last values obtained should be one usable to determine the desired output.

ACTIVATION FUNCTIONS: An activation function is a non-linear function applied by a neuron to introduce non-linear properties in the network. Here we use two nonlinear activation functions they are Hyperbolic Tangent Activation Function (Tanh) and Rectified Linear Unit (ReLU) Activation Function.

- **Hyperbolic Tangent Activation Function (Tanh):** The range of the tanh function is from (-1 to 1). tanh is also sigmoidal (s - shaped). The advantage is that the negative inputs will be mapped strongly negative and the zero inputs will be mapped near zero in the tanh graph. The function is differentiable and monotonic while its derivative is not monotonic.

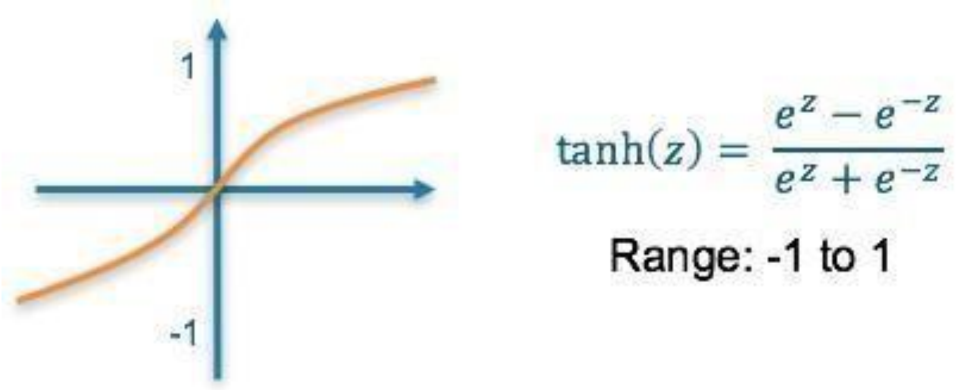


Figure 4: Tanh activation function graph

- **Rectified Linear Unit (ReLU) Activation Function :** The ReLU is the most used activation function in the world right now. Since, it is used in almost all the convolutional neural networks or deep learning. The function and its derivative both are monotonic.

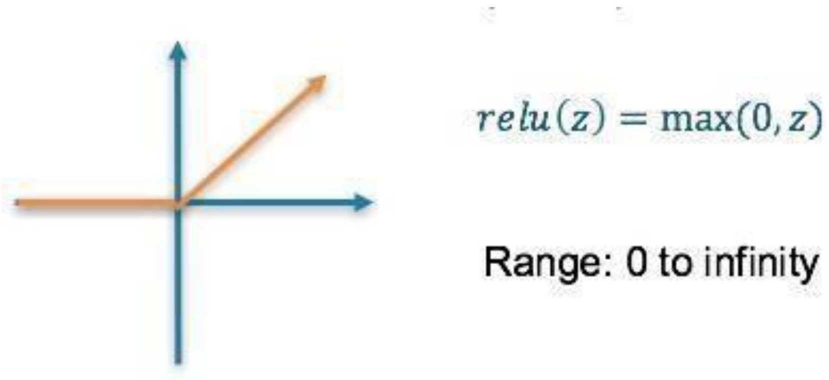


Figure 5: Relu activation function graph

MULTI-LAYER PERCEPTRON (MLP) : A multilayer perceptron (MLP) is a class of feedforward artificial neural network (ANN). The term MLP is used ambiguously, sometimes loosely to refer to any feedforward ANN, sometimes strictly to refer to networks composed of multiple layers of perceptron (with threshold activation).

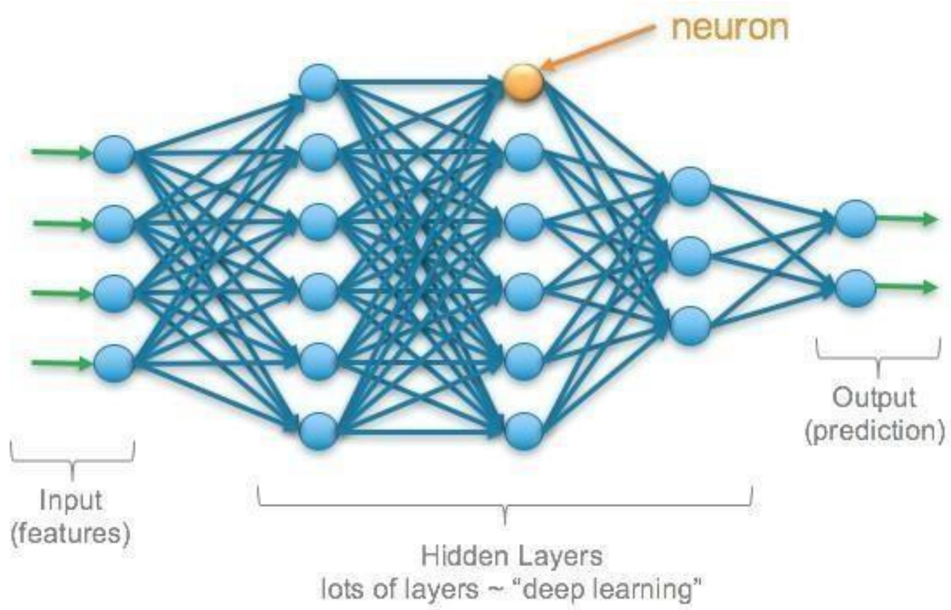
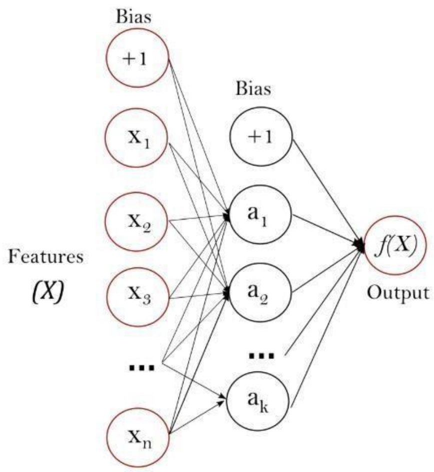


Figure 6: MLP diagram

An MLP must have at least three layers: the input layer, a hidden layer and the output layer. They are fully connected; each node in one layer connects with a weight to every node in the next layer. Multi-layer Perceptron (MLP) is a supervised learning algorithm that learns a function f(.):Rm→Ro by training on a dataset, where m is the number of dimensions for input and o is the number of dimensions for output. Given a set of features X=x1,x2,...,xm and a target y, it can learn a non-linear function approximator for either classification or regression. It is different from logistic regression, in that between the input and the output layer, there can be one or more non-linear layers, called hidden layers. The figure below shows a one hidden layer MLP with scalar output.



The leftmost layer, known as the input layer, consists of a set of neurons {xi|x1,x2…xm} representing the input features. Each neuron in the hidden layer transforms the values from the previous layer with a weighted linear summation w1x1+w2x2+…wmxm , followed by a non-linear activation function g(.): R− −> R - like the hyperbolic tan function. The output layer receives the values from the last hidden layer and transforms them into output values. The module contains the public attributes coefs and intercepts. Coefs is a list of weight matrices, where weight matrix at index represents the weights between layer i and layer i+1. intercepts_ is a list of bias vectors, where the vector at index i represents the bias values added to layer i+1.

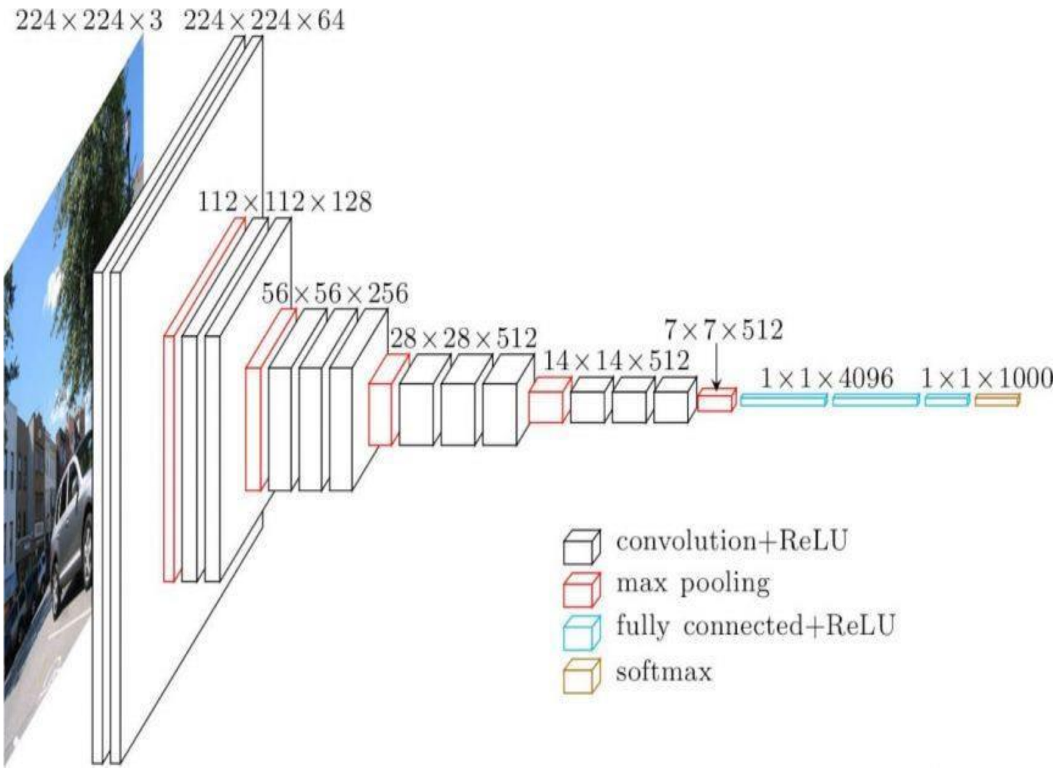
5.vgg16: VGG16 is a convolutional neural network model proposed by K. Simonyan and A.

Zisserman from the University of Oxford in the paper "Very Deep Convolutional Networks for

Large-Scale Image Recognition”. The model achieves 92.7% top-5 test accuracy in

ImageNet, which is a dataset of over 14 million images belonging to 1000 classes. It was one of the famous model submitted to ILSVRC-2014. It makes the improvement over AlexNet by replacing large kernel-sized filters (11 and 5 in the first and second convolutional layer, respectively) with multiple 3×3 kernel-sized filters one after another. VGG16 was trained for weeks and was using

NVIDIA Titan Black GPU’s.



1. PACKAGES :

- **NumPy:**

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed. In this project NumPy has been used to hold the image as a multidimensional – array to be passed as an input to the model designed.

- **Keras:**

Keras is an open source neural network library written in Python. It is capable of running on top of TensorFlow, Microsoft Cognitive Toolkit, or Theano. Designed to enable fast experimentation with deep neural networks, it focuses on being user-friendly, modular, and extensible.

- **Matplotlib:**

Matplotlib is a collection of command style functions that make matplotlib work like MATLAB. Each pyplot function makes some change to a figure: e.g., creates a figure, creates a plotting area in a figure, plots some lines in a plotting area, decorates the plot with labels, etc. In matplotlib.pyplot various states are preserved across function calls, so that it keeps track of things like the current figure and plotting area, and the plotting functions are directed to the current axes.

- **Sklearn:**

Scikit-learn is an open source Python library that implements a range of machine learning, preprocessing, cross-validation and visualization algorithms using a unified interface. The important features of scikit-learn are it is simple and efficient tool for data mining and analysis. It features various classification, regression and clustering algorithms including support vector machines, random forests, gradient boosting, kmeans, etc.

- **Tkinter:**

It is a Python binding to the Tk GUI toolkit. It is the standard Python interface to the Tk GUI toolkit, and is Python's de facto standard GUI. Tkinter is included with standard Linux, Microsoft Windows and Mac OS X installs of Python. As with most other modern Tk bindings, Tkinter is implemented as a Python wrapper around a complete Tcl interpreter embedded in the Python interpreter. Tkinter calls are translated into Tcl commands which are fed to this embedded interpreter, thus making it possible to mix Python and Tcl in a single application.

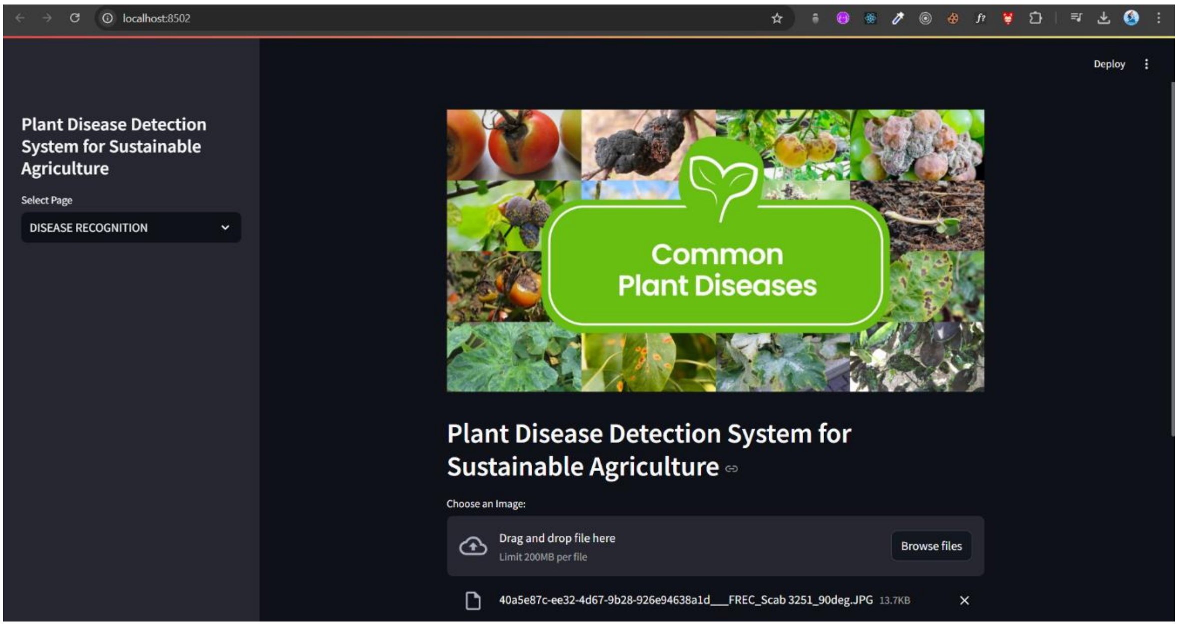
- **Tensorflow:**

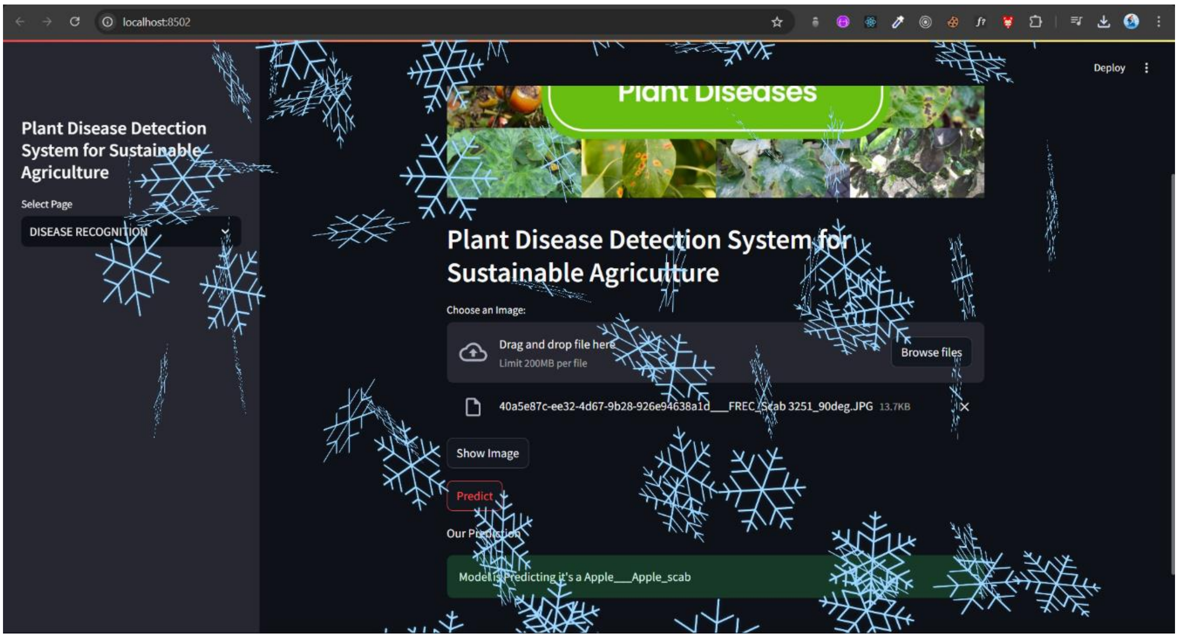
A free and opensource software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. Its flexible architecture allows for the easy deployment of computation across a variety of platforms (CPUs, GPUs, TPUs), and from desktops to clusters of servers to mobile and edge devices. TensorFlow computations are expressed as stateful dataflow graphs. The name TensorFlow derives from the operations that such neural networks perform on multidimensional data arrays, which are referred to as *tensors*. During the Google I/O conference in June 2016, Jeff Dean stated that 1,500 repositories on GitHub mentioned TensorFlow, of which only 5 were from Google.

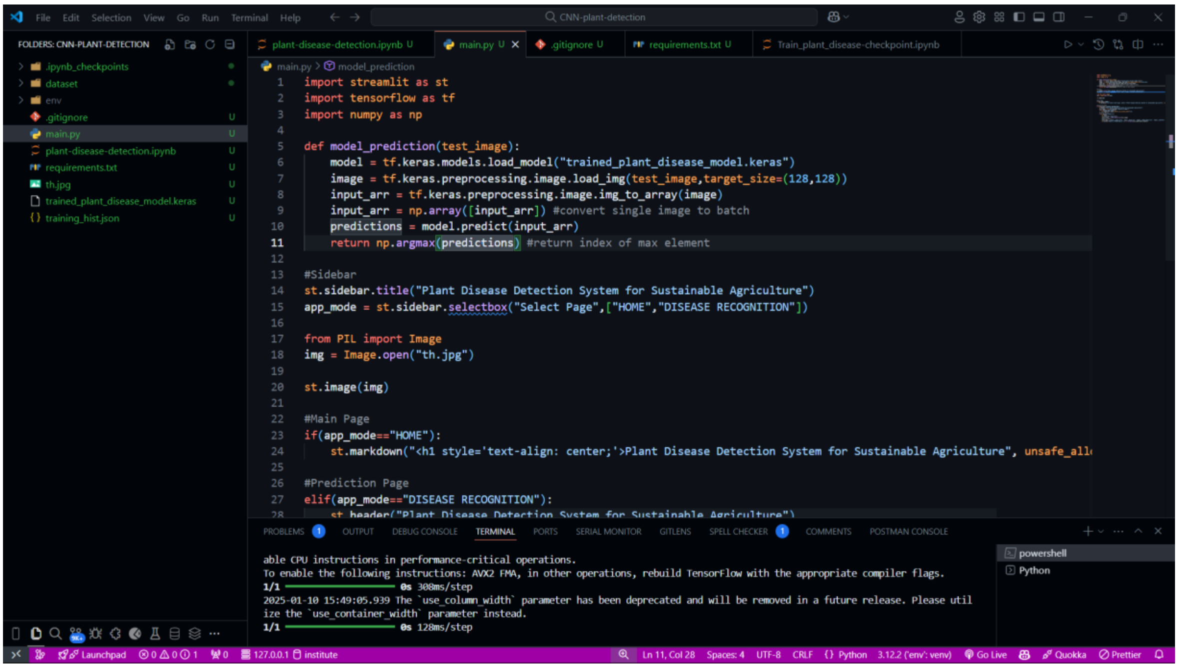
CHAPTER 4

Implementation and Result

4.1 Snap Shots of Result:







4.2 GitHub Link for Code:

https://github.com/neerajkhatri04/Plant-Disease-Detection-System-for-Sustainable-Agriculture

CHAPTER 5

Discussion and Conclusion

5.1 Future Work

1. **Expand the Dataset**:
   - Include images of more plant species and additional diseases to improve the model's versatility and generalizability.
   - Incorporate images with varying environmental conditions (e.g., lighting, background clutter) to enhance robustness.

2. **Enhance Model Performance**:
   - Explore advanced architectures such as EfficientNet or Vision Transformers (ViT) for better accuracy and computational efficiency.
   - Implement transfer learning to leverage pre-trained models for faster and more efficient training.

3. **Real-time Implementation**:
   - Develop mobile and web applications with real-time disease detection capabilities to make the system accessible to farmers globally.

4. **Integrate Multimodal Data**:
   - Combine leaf image data with additional inputs such as soil quality, weather data, and crop growth stage for more holistic disease diagnostics.

5. **Explainable AI**:
   - Incorporate explainable AI techniques to provide visual heatmaps or feature importance, helping users understand how the system identifies diseases.

6. **Support Multilingual Accessibility**:
   - Develop multilingual interfaces to cater to diverse agricultural communities worldwide.

7. **Sustainability Focus**:
   - Integrate recommendations for eco-friendly treatments or pest control measures to promote sustainable farming practices.

5.2 Conclusion

This project successfully demonstrates an AI-powered solution for detecting plant diseases, addressing a critical challenge in agriculture. By employing a CNN-based model, the system achieves high accuracy in diagnosing diseases, enabling farmers to take timely actions to protect their crops. The project contributes significantly to sustainable agriculture by reducing crop losses, optimizing pesticide usage, and supporting food security.

The developed system not only highlights the potential of deep learning in addressing real-world challenges but also lays the groundwork for future innovations in agricultural technology. With further enhancements, such as real-time deployment and integration with additional data sources, this solution can become a vital tool in the global effort toward sustainable and efficient farming.

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